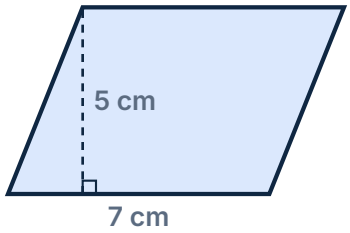
Area of Triangles and Other Shapes

Move a piece or double the shape: make a rectangle.

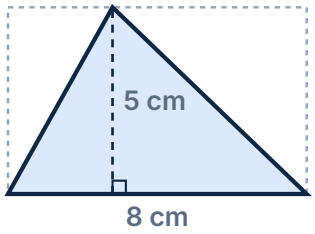
A parallelogram: cut off a triangle, move it over



The triangle on the left fits on the right. Now it is a rectangle, 7 by 5.

$7 \times 5 = 35$ sq cm

A triangle is half of a rectangle

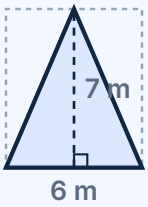


The dashed rectangle is 8 by 5 = 40. The triangle is half of it.

$8 \times 5 \div 2 = 20$ sq cm

Other shapes: split them into rectangles and triangles, then add

Your turn!



$\boxed{6} \times \boxed{7} \div 2 = \boxed{21}$ sq m

For teachers

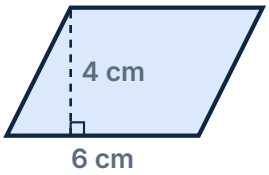
Goal: area of parallelograms, triangles, trapezoids and shapes made of them (6.G.1). Say “What rectangle is hiding here?” The height always meets the base in a square corner.

Area of Triangles and Other Shapes: Practice

Find the rectangle. Then base × height.

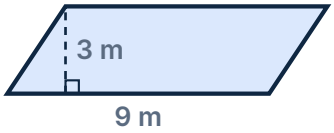
Part 1 Parallelograms Move the triangle: it is a rectangle.

1)



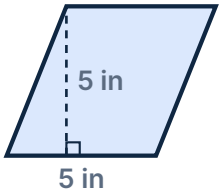
6 × 4 = 24 sq cm

2)



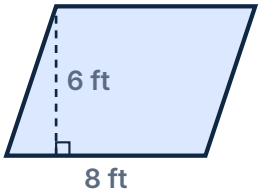
9 × 3 = 27 sq m

3)



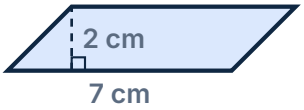
5 × 5 = 25 sq in

4)



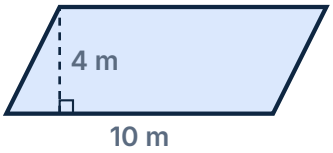
8 × 6 = 48 sq ft

5)



7 × 2 = 14 sq cm

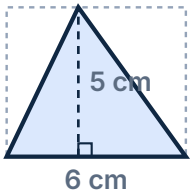
6)



10 × 4 = 40 sq m

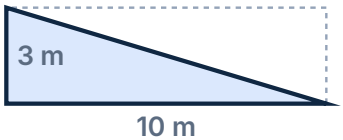
Part 2 Triangles Half of the dashed rectangle.

7)



6 × 5 ÷ 2 = 15 sq cm

8)

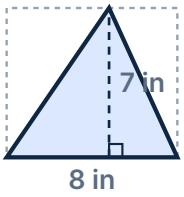


10 × 3 ÷ 2 = 15 sq m

Area of Triangles and Other Shapes: Practice (continued)

Part 2 Triangles (continued)

9)



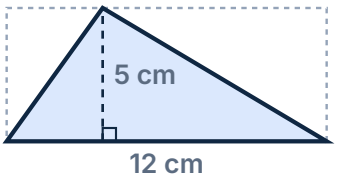
8 × 7 ÷ 2 = 28 sq in

10)



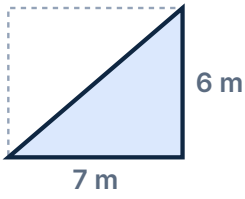
4 × 9 ÷ 2 = 18 sq ft

11)



12 × 5 ÷ 2 = 30 sq cm

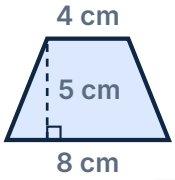
12)



7 × 6 ÷ 2 = 21 sq m

Part 3 Trapezoids Top + bottom, times the height, then half.

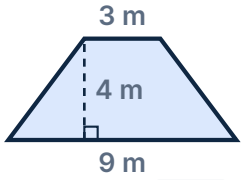
13)



4 + 8 = 12

12 × 5 ÷ 2 = 30 sq cm

14)



3 + 9 = 12

12 × 4 ÷ 2 = 24 sq m

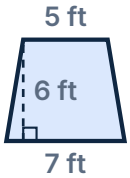
15)



6 + 10 = 16

16 × 3 ÷ 2 = 24 sq in

16)



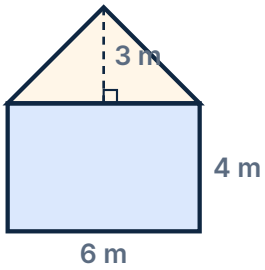
5 + 7 = 12

12 × 6 ÷ 2 = 36 sq ft

Area of Triangles and Other Shapes: Practice (continued)

Part 4 Put shapes together A rectangle and a triangle: find each, then add.

17)

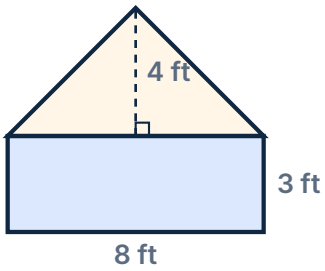


rectangle: 24

triangle: 9

24 + 9 = 33 sq m

18)

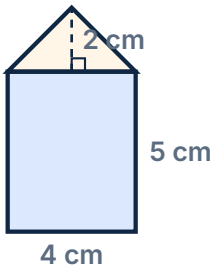


rectangle: 24

triangle: 16

24 + 16 = 40 sq ft

19)

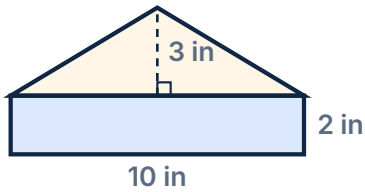


rectangle: 20

triangle: 4

20 + 4 = 24 sq cm

20)



rectangle: 20

triangle: 15

20 + 15 = 35 sq in

Area of Triangles and Other Shapes: Practice (continued)

Part 5 Triangles No help this time.

21) triangle, base 14 cm, height 5 cm: area = 35 sq cm

22) triangle, base 9 m, height 8 m: area = 36 sq m

23) triangle, base 11 in, height 6 in: area = 33 sq in

24) triangle, base 16 ft, height 3 ft: area = 24 sq ft

25) triangle, base 7 cm, height 12 cm: area = 42 sq cm

26) triangle, base 20 m, height 9 m: area = 90 sq m

27) triangle, base 13 in, height 4 in: area = 26 sq in

28) triangle, base 15 cm, height 6 cm: area = 45 sq cm

Part 6 Parallelograms No help this time.

29) parallelogram, base 12 cm, height 7 cm: area = 84 sq cm

30) parallelogram, base 9 m, height 9 m: area = 81 sq m

31) parallelogram, base 15 ft, height 4 ft: area = 60 sq ft

Area of Triangles and Other Shapes: Practice (continued)

Part 6 Parallelograms (continued)

32) parallelogram, base 8 in, height 11 in: area = 88 sq in

33) parallelogram, base 14 m, height 6 m: area = 84 sq m

34) parallelogram, base 25 cm, height 3 cm: area = 75 sq cm

Part 7 Trapezoids No help this time.

35) trapezoid, sides 5 and 11 cm, height 4 cm: area = 32 sq cm

36) trapezoid, sides 8 and 12 m, height 5 m: area = 50 sq m

37) trapezoid, sides 3 and 7 in, height 6 in: area = 30 sq in

38) trapezoid, sides 9 and 15 ft, height 2 ft: area = 24 sq ft

39) trapezoid, sides 6 and 14 cm, height 7 cm: area = 70 sq cm

40) trapezoid, sides 10 and 16 m, height 3 m: area = 39 sq m

Part 8 Find the height Area × 2 ÷ base.

41) triangle, area 28 sq cm, base 8 cm: height = 7 cm

Area of Triangles and Other Shapes: Practice (continued)

Part 8 Find the height (continued)

42) triangle, area 45 sq m, base 10 m: height = 9 m

43) triangle, area 33 sq in, base 6 in: height = 11 in

44) triangle, area 24 sq ft, base 12 ft: height = 4 ft

45) triangle, area 42 sq cm, base 14 cm: height = 6 cm

46) triangle, area 20 sq m, base 5 m: height = 8 m

Part 9 More triangles

47) triangle, base 6 in, height 5 in: area = 15 sq in

48) triangle, base 4 m, height 5 m: area = 10 sq m

49) triangle, base 22 ft, height 10 ft: area = 110 sq ft

50) triangle, base 10 m, height 9 m: area = 45 sq m

51) triangle, base 3 m, height 18 m: area = 27 sq m

52) triangle, base 24 m, height 14 m: area = 168 sq m

Area of Triangles and Other Shapes: Practice (continued)

Part 9 More triangles (continued)

53) triangle, base 22 cm, height 12 cm: area = 132 sq cm

54) triangle, base 24 ft, height 19 ft: area = 228 sq ft

55) triangle, base 21 ft, height 18 ft: area = 189 sq ft

56) triangle, base 8 cm, height 3 cm: area = 12 sq cm

57) triangle, base 8 cm, height 17 cm: area = 68 sq cm

58) triangle, base 21 m, height 14 m: area = 147 sq m

Part 10 More parallelograms

59) parallelogram, base 22 cm, height 12 cm: area = 264 sq cm

60) parallelogram, base 13 m, height 5 m: area = 65 sq m

61) parallelogram, base 5 cm, height 11 cm: area = 55 sq cm

62) parallelogram, base 20 m, height 9 m: area = 180 sq m

63) parallelogram, base 10 ft, height 5 ft: area = 50 sq ft

Area of Triangles and Other Shapes: Practice (continued)

Part 10 More parallelograms (continued)

64) parallelogram, base 4 cm, height 4 cm: area = 16 sq cm

65) parallelogram, base 12 m, height 2 m: area = 24 sq m

66) parallelogram, base 4 ft, height 2 ft: area = 8 sq ft

Part 11 More trapezoids

67) trapezoid, sides 5 and 17 cm, height 6 cm: area = 66 sq cm

68) trapezoid, sides 6 and 9 in, height 2 in: area = 15 sq in

69) trapezoid, sides 7 and 17 ft, height 7 ft: area = 84 sq ft

70) trapezoid, sides 3 and 7 ft, height 3 ft: area = 15 sq ft

71) trapezoid, sides 6 and 9 in, height 12 in: area = 90 sq in

72) trapezoid, sides 3 and 12 cm, height 4 cm: area = 30 sq cm

73) trapezoid, sides 8 and 15 m, height 6 m: area = 69 sq m

74) trapezoid, sides 12 and 20 m, height 10 m: area = 160 sq m

Area of Triangles and Other Shapes: Practice (continued)

Part 12 More heights

75) triangle, area 78 sq ft, base 12 ft: height = 13 ft

76) triangle, area 36 sq ft, base 6 ft: height = 12 ft

77) triangle, area 30 sq cm, base 5 cm: height = 12 cm

78) triangle, area 48 sq m, base 12 m: height = 8 m

79) triangle, area 24 sq m, base 6 m: height = 8 m

80) triangle, area 24 sq in, base 4 in: height = 12 in

81) triangle, area 42 sq ft, base 7 ft: height = 12 ft

82) triangle, area 16 sq cm, base 4 cm: height = 8 cm

Area of Triangles and Other Shapes: Practice (continued)

Part 13 Remember? Problems from earlier packets.

83) $x + 18 = 45$: $x =$ **27**

84) $6(a + 4) = 6a +$ **24**

85) $|-13| =$ **13**

86) gcf of 18 and 45 = **9**

87) $4.2 \times 1.5 =$ **6.3**

88) $12 : 16 =$ **3** : **4**

Part 14 Word problems Draw the shape. Find the rectangle.

89) A sail is a triangle with a base of 6 m and a height of 9 m. What is its area?

Number sentence **6 × 9 ÷ 2 = 27**

Answer **27** sq m

Draw or show your work

90) A garden is a parallelogram, base 12 ft and height 7 ft. How much ground does it cover?

Number sentence **12 × 7 = 84**

Answer **84** sq ft

Draw or show your work

91) A ramp's side is a trapezoid: sides 4 ft and 10 ft, height 3 ft. What is its area?

Number sentence **14 × 3 ÷ 2 = 21**

Answer **21** sq ft

Draw or show your work

Area of Triangles and Other Shapes: Practice (continued)

Part 14 Word problems (continued)

92) A flag is a triangle with area 30 sq in and base 10 in. How tall is it?

Number sentence 30 × 2 ÷ 10 = 6

Answer 6 in

Draw or show your work

93) Challenge: A wall 8 m wide and 3 m tall has a triangle on top 2 m tall. What is its area?

Number sentence 24 + 8 = 32

Answer 32 sq m

Draw or show your work

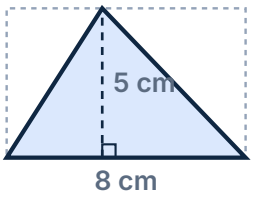
Check-In: Area of Triangles and Other Shapes

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

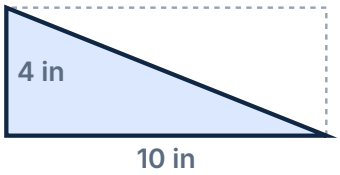
Part 1 Triangles

1)



8 × 5 ÷ 2 = 20 sq cm

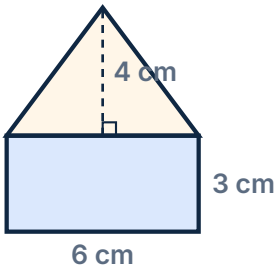
2)



10 × 4 ÷ 2 = 20 sq in

Part 2 Put shapes together

3)



rectangle: 18

triangle: 12

18 + 12 = 30 sq cm

Part 3 Find the area

4) triangle, base 10 ft, height 11 ft: area = 55 sq ft

5) trapezoid, sides 7 and 9 cm, height 5 cm: area = 40 sq cm

6) parallelogram, base 13 m, height 4 m: area = 52 sq m

7) triangle, base 18 cm, height 5 cm: area = 45 sq cm